

## LA ROCHELLE LE BOUT BLANC, France

WMO: 073150

Lat: 46.1526N

Lon: 1.1587W

Elev: 33

StdP: 14.68

Time Zone: 1.00 (EUC)

Period: 94-17

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 27.7       | 30.4 | 20.3                          | 15.2 | 30.6 | 23.6 | 17.9 | 33.3 | 29.5                  | 51.9 | 25.8 | 52.0 | 7.6                   | 80   | 0.508 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 12.2                   | 85.9            | 68.3 | 81.9 | 67.2 | 78.6 | 66.4 | 71.4                | 80.6 | 69.8 | 78.0 | 68.4 | 75.5 | 8.4                  | 90   |

| Dehumidification DP/MCDB and HR |       |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 68.5                            | 105.0 | 75.3 | 67.0 | 99.3 | 73.5 | 65.7 | 94.9 | 72.1 | 35.1          | 81.0 | 33.8 | 78.2 | 32.6 | 75.9 | 78.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |       |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |       |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |       |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |
| 24.2              | 20.8 | 18.2 | DB                         | 24.4 | 93.8               | 3.8 | 4.0                                                | 21.6 | 96.7       | 19.4 | 99.0       | 17.3 | 101.3      | 14.5 | 104.2 |
|                   |      |      | WB                         | 22.9 | 74.1               | 3.8 | 1.8                                                | 20.2 | 75.4       | 18.0 | 76.4       | 15.9 | 77.5       | 13.2 | 78.8  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 56.3        | 44.3  | 45.4  | 49.8  | 53.7  | 59.8  | 65.7  | 68.6  | 68.6  | 64.2  | 58.9  | 50.9  | 45.2  | (6)  |
|                                                                        | DBStd       | 10.35       | 6.48  | 5.91  | 5.26  | 4.96  | 5.02  | 5.02  | 4.78  | 4.41  | 4.54  | 5.28  | 5.81  | 6.84  | (7)  |
|                                                                        | HDD50       | 653         | 189   | 142   | 67    | 18    | 0     | 0     | 0     | 0     | 0     | 6     | 57    | 174   | (8)  |
|                                                                        | HDD65       | 3552        | 642   | 550   | 470   | 339   | 177   | 49    | 15    | 10    | 67    | 196   | 423   | 614   | (9)  |
|                                                                        | CDD50       | 2959        | 12    | 12    | 61    | 130   | 305   | 470   | 575   | 577   | 426   | 282   | 84    | 25    | (10) |
|                                                                        | CDD65       | 383         | 0     | 0     | 0     | 1     | 17    | 69    | 125   | 122   | 42    | 7     | 0     | 0     | (11) |
|                                                                        | CDH74       | 2046        | 0     | 0     | 0     | 9     | 117   | 424   | 679   | 591   | 210   | 16    | 0     | 0     | (12) |
|                                                                        | CDH80       | 555         | 0     | 0     | 0     | 0     | 17    | 119   | 208   | 165   | 45    | 1     | 0     | 0     | (13) |
| (14) Wind                                                              | WSAvg       | 8.9         | 9.0   | 9.5   | 9.3   | 9.5   | 9.4   | 9.1   | 9.4   | 8.7   | 8.1   | 7.7   | 8.6   | 8.9   | (14) |
| (15) Precipitation                                                     | PrecAvg     | 28.9        | 2.9   | 2.2   | 2.0   | 2.4   | 2.0   | 1.4   | 1.7   | 1.9   | 2.2   | 3.3   | 3.8   | 3.1   | (15) |
|                                                                        | PrecMax     | 40.0        | 6.5   | 4.6   | 6.4   | 9.4   | 3.5   | 3.3   | 5.1   | 4.3   | 8.3   | 6.8   | 10.2  | 5.8   | (16) |
|                                                                        | PrecMin     | 20.6        | 0.4   | 0.0   | 0.2   | 0.6   | 0.6   | 0.2   | 0.2   | 0.2   | 0.0   | 0.4   | 0.7   | 0.6   | (17) |
|                                                                        | PrecStd     | 5.2         | 1.6   | 1.3   | 1.4   | 1.8   | 0.8   | 0.7   | 0.9   | 1.3   | 1.9   | 1.7   | 2.0   | 1.5   | (18) |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 56.7  | 59.5  | 70.0  | 74.9  | 82.2  | 90.0  | 90.9  | 90.0  | 85.5  | 76.3  | 65.7  | 58.5  | (19) |
|                                                                        |             | MCWB        | 53.1  | 51.8  | 56.0  | 60.6  | 64.5  | 68.9  | 70.6  | 70.4  | 66.4  | 64.4  | 59.2  | 55.5  | (20) |
|                                                                        | 2%          | DB          | 54.9  | 55.9  | 64.6  | 69.1  | 77.3  | 82.6  | 85.6  | 84.7  | 79.2  | 71.6  | 62.1  | 56.7  | (21) |
|                                                                        |             | MCWB        | 52.3  | 50.6  | 54.1  | 57.2  | 63.5  | 66.7  | 68.2  | 69.1  | 65.1  | 63.4  | 57.8  | 54.0  | (22) |
|                                                                        | 5%          | DB          | 53.5  | 53.9  | 60.6  | 65.9  | 72.8  | 78.3  | 81.2  | 80.4  | 75.6  | 68.5  | 60.2  | 55.4  | (23) |
|                                                                        |             | MCWB        | 51.2  | 50.5  | 52.5  | 55.8  | 61.9  | 65.8  | 67.3  | 68.4  | 63.9  | 62.3  | 56.9  | 53.0  | (24) |
|                                                                        | 10%         | DB          | 52.3  | 52.5  | 57.3  | 62.4  | 68.8  | 74.9  | 77.2  | 76.8  | 72.3  | 66.2  | 58.6  | 54.0  | (25) |
|                                                                        |             | MCWB        | 50.2  | 49.9  | 51.3  | 54.6  | 60.2  | 64.4  | 66.6  | 67.4  | 63.4  | 61.4  | 55.6  | 51.9  | (26) |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 54.4  | 53.9  | 57.5  | 61.9  | 68.2  | 71.9  | 72.5  | 74.8  | 70.4  | 66.8  | 60.8  | 56.5  | (27) |
|                                                                        |             | MCDB        | 55.6  | 56.2  | 66.6  | 70.2  | 78.6  | 83.7  | 83.2  | 83.5  | 78.8  | 71.9  | 63.3  | 57.7  | (28) |
|                                                                        | 2%          | WB          | 52.9  | 52.4  | 55.5  | 59.1  | 64.8  | 69.5  | 70.7  | 71.9  | 68.2  | 64.9  | 59.2  | 54.7  | (29) |
|                                                                        |             | MCDB        | 54.3  | 54.4  | 61.7  | 66.4  | 73.8  | 79.1  | 80.5  | 79.8  | 74.3  | 69.0  | 61.2  | 56.1  | (30) |
|                                                                        | 5%          | WB          | 51.8  | 51.3  | 53.9  | 57.0  | 62.7  | 67.6  | 69.2  | 70.1  | 66.5  | 63.5  | 57.5  | 53.4  | (31) |
|                                                                        |             | MCDB        | 53.1  | 53.1  | 58.3  | 63.4  | 70.6  | 75.7  | 77.4  | 77.4  | 72.2  | 67.2  | 59.7  | 54.9  | (32) |
|                                                                        | 10%         | WB          | 50.6  | 50.3  | 52.7  | 55.3  | 60.9  | 65.7  | 67.8  | 68.4  | 65.0  | 62.0  | 55.9  | 52.2  | (33) |
|                                                                        |             | MCDB        | 52.1  | 52.3  | 56.2  | 60.6  | 67.4  | 72.7  | 75.2  | 74.5  | 70.5  | 65.6  | 58.5  | 53.9  | (34) |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 8.5   | 9.5   | 11.7  | 12.4  | 11.9  | 12.5  | 11.8  | 12.2  | 13.4  | 11.5  | 9.5   | 8.9   | (35) |
|                                                                        |             | MCDBR       | 7.8   | 11.2  | 17.9  | 19.3  | 20.5  | 20.8  | 21.0  | 20.1  | 20.1  | 14.7  | 10.6  | 8.2   | (36) |
|                                                                        | 5% WB       | MCWBR       | 6.5   | 7.9   | 10.4  | 10.6  | 10.3  | 9.8   | 9.3   | 9.6   | 9.9   | 8.6   | 7.4   | 6.8   | (37) |
|                                                                        |             | MCWBR       | 7.0   | 8.6   | 13.2  | 16.1  | 16.7  | 17.3  | 15.8  | 15.1  | 14.6  | 11.8  | 9.3   | 7.3   | (38) |
| (40) Clear-Sky Solar Irradiance                                        | taub        | taub        | 0.330 | 0.344 | 0.373 | 0.394 | 0.399 | 0.410 | 0.401 | 0.392 | 0.370 | 0.368 | 0.342 | 0.326 | (40) |
|                                                                        |             | taud        | 2.459 | 2.411 | 2.336 | 2.282 | 2.323 | 2.323 | 2.348 | 2.391 | 2.429 | 2.440 | 2.450 | 2.444 | (41) |
|                                                                        | Ebn at Noon | Ebn at Noon | 243   | 262   | 270   | 273   | 276   | 272   | 273   | 272   | 268   | 250   | 235   | 230   | (42) |
|                                                                        |             | Edh at Noon | 22    | 28    | 34    | 39    | 39    | 39    | 38    | 35    | 31    | 27    | 22    | 20    | (43) |
| (44) All-Sky Solar Radiation                                           | RadAvg      | RadAvg      | 384   | 688   | 1063  | 1507  | 1767  | 1978  | 1912  | 1693  | 1343  | 807   | 465   | 340   | (44) |
|                                                                        | RadStd      | RadStd      | 38    | 70    | 119   | 144   | 136   | 151   | 93    | 121   | 88    | 69    | 49    | 41    | (45) |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| Regional (19 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
|                         | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page